Zane Perry

Zane.L.Perry@gmail.com | (720) - 440 - 4783 | Boulder, CO
www.zane-perry.com | www.linkedin.com/in/zane-perry | www.github.com/zane-perry

Objective: Computational applied mathematician seeking roles in signal analysis, machine learning, and data science developing robust algorithms to extract structure from real-world signals.

EDUCATION

University of Colorado Boulder — Boulder, CO

M.S. Applied Mathematics — August 2025 - Expected May 2026

Applied Data Science & Stochastic Processes, Machine Learning; Cumulative GPA: 4.0

B.S. in Computer Science, B.S. in Applied Mathematics — August 2021 - May 2025

Summa Cum Laude, Dean's List Fall 2021- Spring 2025, Cumulative GPA: 4.0

Skills

Languages: Python, Java, TypeScript/JavaScript, C/C++, Assembly, MATLAB, Bash, OOP
Signal/DSP: Fourier Analysis, Time Series Analysis, Filtering/Denoising, Wavelets, Feature Extraction, DSP
Tools: Numpy, Scipy, PyTorch, Pandas, Sci-Kit Learn, CUDA, torchaudio
ML/AI: Supervised, Unsupervised, Deep Learning, Generative AI, Multimodal Models, Physics-Informed
Relevant Coursework: Fourier Analysis, Music Technology, Time Series & Signal Analysis, Numerical Analysis

WORK EXPERIENCE

CU Electrical Engineering Department - Genetic Logic Lab — Boulder, CO

Research Assistant, Software Developer — June 2022 – Present

- Design, Implement, and Test frontend/backend frameworks and endpoints using React, Spring, Java, and Javascript for a Genetic info repository storing 100k+ genetic designs, enhancing data accessibility in response to user feedback.
- Execute full stack development based on specified user requirements. Manage deployment on Azure and Kubernetes.
- Mentor an undergraduate student on the software development life cycle.

Handshake MOVE Fellowship — Remote

AI Trainer - Math Expert, Reviewer (Contract) — May 2025 – December 2025

- Developed 100 complex mathematical prompts to assess and improve the performance of large language models.
- Analyzed LLM outputs for mathematical rigor, enhancing model complexity and outputs.
- Audited other fellows prompts and responses to address complexity, analyze reasoning, and suggest improvements.

Space Environment Technologies — Denver, CO

Radiation Division Computer Science Intern — May 2024 – August 2024

- Automated data upload/analysis processes with APIs and file systems, to reduce manual processing time by 95%.
- Worked with specialized atmospheric radiation models in order to enhance understanding of space radiation effects.
- Designed and built Agile front-end user apps for flight planning and radiation risk analysis.

CU Computer Science Department

Boulder, CO

Data Science Course Assistant for 300 students — August 2023 - December 2023

- Reviewed homework assignments and lecture topics to recommend improvements for the benefit of student learning.
- Held office hours to review concepts, including practical coding applications and theoretical mathematical derivations.

AWARDS AND LEADERSHIP

- National Merit Scholar, 2021.
- University of Colorado, Boulder Chancellor's Recognition Award, Spring 2025.
- President, Theta Tau Professional Engineering Fraternity Eta Gamma Chapter, (100 members, 300 alumni), 2022-2025.
- Chair/Webmaster, Engineering Excellence Fund, (responsible for $500k in engineering grants), 2021-2023.
- Writer/Webmaster, Colorado Engineering Magazine, (Distributed to 40k students/alumni), 2021-2025.

TECHNICAL PROJECT PORTFOLIO – UNIVERSITY OF COLORADO, BOULDER

Synbiohub 2/3, A Standards Based Genetic Design Repository
Spring 2022 - Present

- **Motivation:** Modernize a tightly coupled, legacy genetic design repository that struggled to scale and was difficult to evolve alongside SBOL and triplestore standards and modern web UX expectations.
- **Approach:** Built a modular plugin interface to decouple extensions from the core platform, and designed a cleaner frontend/ backend separation around REST APIs.
- **Tech and Methods:** React/Redux, Java/Spring, REST APIs, PostgreSQL & SPARQL, Docker/Kubernetes/Azure.

Constraint Encoding in Physics-Informed Neural Network PDE Solvers
April 2026 (Current)

- **Motivation:** Neural networks can estimate PDE solutions far more efficiently than classical solvers, but naive training can produce unstable or non-physical solutions; physics-informed neural networks (PINNs) incorporate PDE-theoretic constraints to improve correctness.
- **Approach:** Analyze and prototype constraint-aware PINN objectives incorporating boundary data, weak formulations, variational structure, and admissibility-style constraints across elliptic/parabolic/hyperbolic PDEs.
- **Tech and Methods:** Deep learning, physics-informed ML, advanced PDE analysis, Numerical Simulation.

Senior Capstone - Festo AI Assisted Rapid Prototyper for DevOps
April 2025

- **Motivation:** Speed up DevOps prototyping by turning uploaded Epics/User Stories into a runnable starter app, reducing time spent on initial scaffolding and wiring.
- **Approach:** Built a React workflow to upload, generate, edit/view, and export code, integrating an LLM-backed Python service and an in-browser dev environment for inspecting and iterating on generated code and file structure. Worked with a group of 6 students through the entire process overseen by a corporate sponsor.
- **Tech and Methods:** React/Redux, AWS DynamoDB, Generative AI, Chain of Thought Prompting.

Advanced Music Recommendation using Deep Learning and Digital Signal Processing
April 2025

- **Motivation:** Explore music-first recommendation (audio content over metadata/behavior graphs) to better handle personalization and cold-start settings where historical signals are sparse.
- **Approach:** Built an end-to-end pipeline that converts raw audio into DSP feature representations (MFCC, STFT, DWT), then trains single-modal and multimodal neural models for binary user–song compatibility.
- **Tech and Methods:** Hybrid CNN/RNN/LSTM, Signal Transforms, Hyperparameter Analysis, PyTorch, Sci-Kit, Torchaudio.

Boundary Integral Equation Ordinary/Partial Differential Equation Solver
April 2024

- **Motivation:** Avoid instability/ill-conditioning from mesh-based derivative discretization by reformulating boundary value problems as integral equations where numerical error is driven primarily by quadrature accuracy.
- **Approach:** Derived Green's-function solutions for 1D Sturm–Liouville BVPs and extended to 2D Laplace problems using single/double-layer potentials, reducing PDE solves to a boundary-only Fredholm integral equation for an unknown density.
- **Tech and Methods:** Numerical Quadrature, Boundary Discretization, Irregular Domains, Dense Solvers, Numpy, Scipy.

Airfoil Fluid Simulation Using Conformal Mapping
April 2023

- **Motivation:** Model lift-generating flow over an airfoil without full CFD by using ideal potential flow + complex analysis to translate simple geometries into realistic wing cross-sections.
- **Approach:** Constructed the complex velocity potential by superimposing uniform/source–sink/doublet/vortex flows, enforced the Kutta condition, then applied the Joukowsky conformal map to transform cylinder flow into a cambered airfoil and compute circulation/lift (Kutta–Joukowsky).
- **Tech and Methods:** Complex Variables, PDEs, Numerical Visualizations, MATLAB.

Signal Noise Reducer Using Numerical Matrix Factoring
December 2023

- **Motivation:** Denoise signals efficiently by extracting a low-rank structure without relying on full SVD, which becomes expensive on large matrices.
- **Approach:** Implemented rank-revealing QR with column pivoting (and compared Gram–Schmidt, Householder, Givens variants) to estimate numerical rank and build low-rank approximations.
- **Tech and Methods:** Subspace Methods, Signal Processing, Python, Numpy, Numerical Linear Algebra.

Image Compression Using Discrete Wavelet Transforms
December 2022

- **Motivation:** Reduce image storage size while preserving perceptual quality by exploiting the fact that most images concentrate information in low-frequency structure.
- **Approach:** Implemented a 2D Haar DWT as an averaging orthogonal transform, then applied coefficient thresholding to induce sparsity and reconstruct the image via the inverse transform; compared against SVD compression as a numerical baseline.
- **Tech and Methods:** MATLAB, Signal Processing, Numerical Linear Algebra, Computer storage.

For Full Project Descriptions and Artifacts visit: https://zane-perry.com/projects